2 **(a)** 4.0 N, downward (or statement must lead clearly) B1

show application and not definition of Newton's third law (e.g. candidate must state that the force exerted on ball A by ball B is equal and opposite to the force exerted on ball B by ball A) B1

(b) **(i)** not conserved B1

(ii) (average) resultant force acting on A (during contact)
$= mg$ contact force = $(0.50 \cdot 9.81)$ $4.0 = 0.905$ N C1

change in momentum of A = $0.905 \cdot 0.50 = 0.4525$ N s C1

$0.50\ (v_f - 2.0) = 0.4525$ C1

$V_f = 2.9$ m s^{-1} A1